Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/25/2022
PI Name:	Chang, Tammy T. M.D., Ph.D.		
Project Title:	Using Organoids to Determine Gravity Effects on Organogenesis and Vasculogenesis		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	94143-2205	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/24/2018	**End Date:**	07/23/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/23/2021 per NSSC information (Ed., 9/10/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K1470		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research aims to use transcriptomic analyses to determine the gravity-sensitive molecular pathways important for organogenesis. Human induced pluripotent stem cell (iPSC) derived liver organoids will be used as an in vitro model of liver development. In Aim 1, the transcriptome of liver organoids developed under Earth's gravity (1g) and simulated microgravity (micro-g) in rotating wall vessels (RWV) will be determined and gravity-sensitive molecular pathways identified. In Aim 2, gravity-dependent pathways identified in Aim 1 will be cross-analyzed with the liver transcriptomes of space-flown and ground control mice curated in GeneLab to determine the pathways that are most robustly regulated by gravity. In Aim 3, based on the transcriptomic analyses performed in Aims 1 and 2, the CRISPR/Cas9 system will be used to generate a gravity mutant iPSC line with altered signaling in a critical gravity-regulated pathway. The transcriptome of the resulting mutant liver organoid developed under simulated micro-g conditions will be compared to the transcriptome of wild-type organoids cultured in 1g and micro-g to determine the biological significance of the pathway. The proposed research aligns with the over-arching goal of the Space Biology Plan to use genomic approaches to investigate the effect of the space environment on life and addresses guiding questions related to how micro-g impacts tissue morphogenesis and stem cell differentiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research findings will help develop strategies to use microgravity as a tool to advance tissue engineering and generate replacement tissues as a treatment for people who have end-stage organ failure on Earth.
Task Progress:	This research project aimed to address high priority recommendations of the Decadal Survey by investigating human induced pluripotent stem cell (iPSC)-derived liver organoids generated in simulated microgravity (micro-g) as an in vitro model of organogenesis and vasculogenesis. Prevailing terrestrial protocols for generating organoids rely on culturing cells on top of Matrigel surfaces, which are incompletely defined basement membrane matrices extracted from Engelbreth–Holm–Swarm mouse sarcomas. To determine the differentiation and function of liver organoids generated in simulated micro-g compared to Earth's 1g, we differentiated human iPSCs into hepatic endoderm, and then induced organoid formation by culturing either on top of Matrigel (1g) or within rotating wall vessels (RWV) that simulated micro-g. Compared to Matrigel organoids, RWV organoids demonstrated higher numbers of cells expressing hepatocyte nuclear factor 4 alpha (HNF4a), a key master transcriptional regulator of hepatocyte function. RNA-seq analysis showed that RWV organoids had significantly higher expression of 256 hepatocyte-specific genes, encompassing a broad spectrum of liver secretory and metabolic functions. RWV organoids maintained for up to 22 days in culture showed significantly greater albumin secretion, and cytochrome P450 1A1 and 3A4 metabolic activity as compared to Matrigel organoids. Three-dimensional structure of organoids is preserved in RWV simulated micro-g over 3 weeks, whereas three-dimensional organization of organoids on Matrigel disintegrated. Co-culture of iPSC-derived hepatic cells and human endothelial umbilical vein cells (HUVECs) in RWVs showed development of web-like networks resembling microvasculture formation, whereas the distribution of HUVECs within hepatic parenchyma is less, even within Matrigel organoids. Our findings indicate that hepatic organoids that self-assemble under simulated micro-g show greater mature functional characteristics than Matrigel organoids. Matrigel-free organoid self-assembly in simulated gravity may be a high-throughput way to generate highly functional iPSC-derived organoids for tissue engineering research and therapeutic applications.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Abstracts for Journals and Proceedings	Sekyi MT, Weng Y, Desai T, Chang TT. "Hepatic organoids and microporous polycaprolactone bioscaffolds in simulated microgravity." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Chang TT. "Tissue engineering in space." Space Medicine and Its Influence on Earth: Out of this World Healthcare. A lecture video series sponsored by the Osher Mini Medical School for the Public, San Francisco, California, February 24-March 31. 2021. Presentation on March 3, 2021. Lecture video archives. University of California San Francisco, Osher Mini Medical School for the Public, San Francisco, California, February 24-March 31. 2021. , Mar-2021
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Abstracts for Journals and Proceedings	Weng Y, Han S, Sekyi MT, Su T, Mattis AN, Chang TT. "Human iPSC-derived liver organoids generated in rotating wall vessel bioreactors demonstrate improved differentiated functions compared to organoids produced on Matrigel." Poster presented at the International Society for Stem Cell Research (ISSCR) Annual Meeting, San Francisco, CA. June 15-18, 2022. International Society for Stem Cell Research (ISSCR) Annual Meeting, San Francisco, CA. June 15-18, 2022. , Jun-2022
Articles in Peer-reviewed Journals	Weng Y, Han S, Sekyi MT, Su T, Mattis AN, Chang TT. "Self-assembled Matrigel-free iPSC-derived liver organoids demonstrate wide-ranging highly-differentiated liver functions." Stem Cells. 2023 Mar 2;41(2):126-139. https://doi.org/10.1093/stmcls/sxac090 ; PubMed PMID: 36573434; PubMed Central PMCID: PMC9982071 , Mar-2023
Patents	63116612. Provisional U.S. patent issued November 2020. Nov-2020 Chang TT and Sekyi M. "Tissue Orb."
Patents	PCT/US21/60170. International patent cooperation treaty issued November 2021. Nov-2021 Chang TT and Sekyi M. "Novel tissue culture system and reduced gravity culture methods for the production of vascularized tissues. "